Beyond the 2013 table: first values for 64 cells of the bipartite degree–diameter problem

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1. What this note covers

$N^b(\Delta, D)$ is the largest number of vertices of a bipartite graph with maximum degree Δ and diameter at most D. The 2013 survey table (Miller–Širáň, DS14v2, as reproduced on the Combinatorics Wiki) lists lower bounds for $\Delta \le 16$ and $D \le 10$. Between October 26 and November 1, 2026, cloud jobs of the lab built and checked graphs for cells outside that table. This note gives one value for each of 64 of them:

- $\Delta = 17, 18, 19, 20$ at $D = 5, \ldots, 10$ (24 cells);
- $D = 11$ at $\Delta = 3, \ldots, 16$ and $D = 12$ at $\Delta = 3, \ldots, 14$ (26 cells);
- $D = 13$ at $\Delta = 3, \ldots, 10$ and $D = 14$ at $\Delta = 3, \ldots, 8$ (14 cells).

The $\Delta = 17..20$ cells at $D = 3$ and 4 were filled in the same week but are not in the table below: at $\Delta = 17$, 18, 20 they are incidence graphs of PG(2, $\Delta-1$) and W($\Delta-1$) (Moore graphs, so optimal), and at $\Delta = 19$ they are the lab's fourth-hat graph (624) and the AS(17) graph of Corollary A1 (10,982) (both in `papers/Ozden_polished_results.pdf`), checked all-pairs by the coordinator on October 26.

Words used below: a hole is a group element (or vertex pair) not yet within the required distance, and the cost of a candidate is its number of holes; far pairs are vertex pairs at distance more than D; an evaluation is one diameter test inside a search; squad cards (r2/e16 and so on) are short runs by Claude Haiku helpers, each trying one idea, directed by an Opus session (`research/squads/LOG.md`).

Each value is a lower bound: one explicit graph, built from files in the repository. All but three were checked exactly by a program that does not share code with the search that found them; the three level graphs at $D = 7$ rest on a theorem (section 4). A literature card of the new-cells-beyond job (`research/squads/beyond/r2/d10/LIT.md` on its branch) looked for published bipartite values in 40 of these 64 cells ($\Delta = 17..20$ at $D = 5..10$, and $D = 11, 12$ at $\Delta = 3..10$; its list of 48 also has the $D = 3, 4$ cells) and found none, but it could not read the Combinatorics Wiki page (HTTP 503) or the tables of some papers. For the other 24 cells no search was made. So we call the values first values for the lab, not records.

Cubic cells: published graphs. The census of cubic edge-transitive graphs (Foster census) lists bipartite cubic graphs of diameter 11 on 1,200 vertices (CAT(1200,1)), 12 on 2,304 (CSS(2304,5)), 13 on 3,360 (CAT(3360,4)) and 14 on 6,272 (CSS(6272,7)), and the census of cubic vertex-transitive graphs one of diameter 11 on 1,250 (lab notes: `research/squads/own-small/newcells/README.md`, where the 1,200-, 1,250- and 2,304-vertex graphs were also checked all-pairs). All are below the lab's values at $D = 11$, 12 (1,474 and 2,368). But at (3,14) a published graph, CSS(6272,7), is larger than the lab's 5,706, and at (3,13) the lab's 3,400 is only 40 above CAT(3360,4). The (3,13) and (3,14) census graphs have not been checked here, and censuses of larger orders were not searched.

For comparison, the bipartite Moore bound is $M^b(\Delta, D) = 2(1 + (\Delta-1) + \ldots + (\Delta-1)^{D-1})$, and for $\Delta \ge 3$, $5 \le D \le 187$, $D \ne 6$, $N^b(\Delta, D) \le M^b(\Delta, D) - 6$ (Feria-Purón and Pineda-Villavicencio, Discrete Appl. Math. 160 (2012); Prop. 6.4 in arXiv 1010.5651v2, the version that was read). Our values are 100 % of the Moore bound at (17,6), (18,6), (20,6), 80 % at (19,6), 24–27 % at $D = 5$, 14–16 % at $D = 7$, 4.5–7.7 % for the orbit-S Haar graphs at $D = 8..14$ (0.8–3.1 % in the four cells stopped at the jobs' size cap), and 7–36 % for the metacyclic Haar graphs and lifts at $\Delta = 3..6$.

2. Where the numbers come from

Every value in the table was copied from one of these, and the vertex counts were re-derived from the group orders or formulas (`python3 research/paper/make_beyond.py` prints the comparison and rewrites the table in this file):

job (date)	branch	cells
new-cells-beyond (Oct 26)	claude/new-cells-beyond-etos3t	Δ = 17..20 at D = 3..10; first values at D = 11, 12 for Δ = 3..10
conics-2 (Oct 29–30)	claude/conics-2-diameter-7-ic0oat	(17,7)
own-small (Oct 26)	claude/own-small-cells-dkwo42	(3,12)
beyond-d11 (Oct 31–Nov 1)	claude/new-cells-d11-d12-beyond-8sas9r	D = 11 at Δ = 11..14, D = 12 at Δ = 11..13; raises at Δ = 4..10
beyond-d11b (Nov 1)	claude/beyond-d11-d12-new-cells-3t3djx	(15,11), (16,11), (14,12); raises of (12,12), (13,12) and of Δ = 4..6
beyond-d13 (Nov 1)	claude/beyond-d13-search-z35z61	D = 13 at Δ = 3..10, D = 14 at Δ = 3..8; raise of (4,12)
beyond-small (Nov 1)	claude/beyond-small-cells-fwb6si	(17,5); exact checks at D = 6

(Dates are when the jobs ran; the harvest and the coordinator recorded the values on master between October 26 and November 1.) On master the values live in `research/RESULTS.md` (harvest and coordinator lines), the harvest check reports `research/frontier/runs/harvest_check_2026-10-31.txt` and `…_2026-11-01.txt`, and the coordinator's logs `research/coordinator_check/check_2026-10-29.txt` … `check_2026-11-01.txt`. RESULTS.md has no line yet for (17,5) 35,618, (3,13), (3,14), (4,12) 74,066, (4,13) or (4,14); on master these are in the coordinator logs of October 31 and November 1, with their edge lists. On the branches they are in `research/frontier/beyond/BEYOND_TABLE.md` (each branch has its own version), `research/frontier/beyond/BEYOND_SMALL.md` (beyond-small), `research/frontier/beyond13/` (beyond-d13) and `research/frontier/beyond11/` (beyond-d11). Master has no BEYOND_TABLE.md yet. Where the branches and master disagree, the table takes the largest value that has a check of its own, and says where it is. The disagreements: (17,5), master 35,618, beyond-small 35,728; (3,11), master 1,344 (a voltage lift from own-small), every BEYOND_TABLE.md 1,474; (3,12), master 2,368, the BEYOND_TABLE.md files 2,176; (17,7), master 5,693,959, the BEYOND_TABLE.md files 5,631,276; (12,12) and (13,12), master and beyond-d11b about 30 billion, the older tables about 20 billion; and (4,11), (5,11), (6,11), (4,12), (5,12), which beyond-d13 and beyond-d11b raised in parallel, where master holds the larger value of each pair.

3. The table

Columns. construction: "Haar, orbit S" is a Haar graph Haar(G, S) with S a union of orbits of a group automorphism (section 5.1); "Haar, random S" has S searched with no imposed symmetry (section 5.2); "(size cap)" marks the four cells where the job stopped at its size limit with the first S that worked. Group or base graph: $Z_m \rtimes_r Z_k$ is the group of pairs (a, j), a mod m, j mod k, with (a, j)(b, l) = (a + r^j b, j + l); "Frobenius" means m is prime and r has order k mod m (so the group has trivial centre). Check: "+" joins two independent checkers in one report (the harvest's or the coordinator's); "," joins checks run by the search session; a semicolon separates checks run at different times by different people:

- gen = `research/frontier/haarcheck_gen.c`; sl = `research/frontier/haarcheck_sl.c`; psl = `research/structured/tools/haarcheck_psl.c` (the reference checkers); h2 = `research/frontier/haarcheck2.c`; ind = `research/frontier/check_haar_indep.py`; bh2 = `research/frontier/build_haar2.py`; mc = `research/frontier/beyond13/metacheck.py` or `metacheck2.py` (beyond-d13 branch). All of these build the group from its definition and run a BFS from (e,A) and from (e,B); see section 4.
- apsp = all-pairs BFS with no symmetry assumed (`research/lifts/coordinator_check/apsp.c`, or its multithreaded versions apsp_mt / apsp_fast); "coord." = run by the coordinator on the Mac mini; "8 ranges" = the source vertices split over eight cloud workers.
- auto = exact BFS from one vertex per orbit of an automorphism group that the checker verifies itself (`research/frontier/beyond/hex6/apsp_auto.c`, beyond-small branch).
- samp = BFS from 64 random vertices only (not a proof); "theorem" = the diameter bound is a theorem for every choice in the construction.
- † = run by this job; logs in `research/paper/beyond_checks/`.

Where: "master" = harvested or recorded by the coordinator on master; "branch" = only on the job's branch. Footnotes: [a] master has 35,618 (coordinator apsp, October 31). [b] no exact check (section 4). [c] master has

1,344. [d] one checker so far (section 4). [e] a published census graph, CSS(6272,7), is larger (section 1). [f] the session's apsp logs are on the beyond-d11b branch (`research/frontier/runs/cloudcheck/d4D11_meta_523_29_9_b2h01.check.txt`, `d5D11_meta_5023_31_2886_b5k05.check.txt`); master's harvest has bh2 only.

(Δ, D)	value	construction	group or base graph	check	where
(17, 5)	35,728	route-chart lift + 1,936 hubs (88 Z_{22}-orbits)	(16,5) graph: $K_{3,3}$ + doubled matching, Z_{22}, fibre F_4[4]	apsp, bfs_all; apsp†	branch [a]
(18, 5)	47,600	multi-doubled-matching lift	W(13); 2 doubled matchings; $Z_2 \times Z_5$	apsp; coord. apsp	master
(19, 5)	52,428	doubled-matching lift	W(16); Z_6	apsp; coord. apsp	master
(20, 5)	75,000	route-chart lift over F_5	$K_{3,3}$ + doubled matching, Z_{20}, fibre F_5[4]	apsp; coord. apsp	master
(17, 6)	2,236,962	Moore graph	H(16)	classical; auto	branch
(18, 6)	3,017,196	Moore graph	H(17)	classical; auto	branch
(19, 6)	3,203,968	Theorem D, q = 17, m = 2	H(17) doubled with levels and whiskers	theorem (Lean); auto	branch
(20, 6)	5,227,320	Moore graph	H(19)	classical; auto	branch
(17, 7)	5,693,959	cover of H(13) with freed incidences + hubs	7 copies of H(13); 32,356 + 30,327 hubs	coord. apsp; apsp (8 ranges)	master
(18, 7)	7,240,212	level graph	9 copies of H(13), 4 matchings, Z_9	theorem; samp	branch [b]
(19, 7)	11,184,810	level graph	5 copies of H(16), 2 matchings, Z_5	theorem; samp	branch [b]
(20, 7)	15,658,734	level graph	7 copies of H(16), 3 matchings, Z_7	theorem; samp	branch [b]
(17, 8)	41,373,072	Haar, orbit S	$SL(2,53) \times Z_{139}$	sl + h2	master
(18, 8)	57,637,680	Haar, orbit S	$SL(2,61) \times Z_{127}$	sl + h2	master
(19, 8)	86,317,296	Haar, orbit S	$PSL(2,103) \times Z_{79}$	psl + ind	master
(20, 8)	122,739,120	Haar, orbit S	$PSL(2,71) \times Z_{343}$	psl + ind	master
(17, 9)	501,707,400	Haar, orbit S	$PSL(2,101) \times Z_{487}$	psl + ind	master
(18, 9)	790,626,816	Haar, orbit S	$SL(2,127) \times Z_{193}$	sl + h2	master
(19, 9)	1,218,394,320	Haar, orbit S	$SL(2,131) \times Z_{271}$	sl + h2	master
(20, 9)	1,823,175,864	Haar, orbit S	$PSL(2,163) \times Z_{421}$	psl + gen	master
(17, 10)	7,022,052,336	Haar, orbit S	$PSL(2,313) \times Z_{229}$	psl + gen	master
(18, 10)	11,787,972,672	Haar, orbit S	$PSL(2,223) \times Z_{1063}$	psl + gen	master
(19, 10)	19,233,722,496	Haar, orbit S	$SL(2,607) \times Z_{43}$	sl + gen	master
(20, 10)	30,518,778,360	Haar, orbit S	$PSL(2,421) \times Z_{409}$	psl (2 halves) + gen	master
(3, 11)	1,474	Haar, all pairs {x, y} searched	$Z_{67} \rtimes_9 Z_{11}$ (Frobenius)	apsp; apsp†	branch [c]
(4, 11)	30,334	Haar, random S	$Z_{523} \rtimes_9 Z_{29}$ (Frobenius)	bh2, apsp [f]; mc†	master
(5, 11)	311,426	Haar, random S	$Z_{5023} \rtimes_{2886} Z_{31}$ (Frobenius)	bh2, apsp [f]; mc†	master
(6, 11)	2,122,518	Haar, random S	$Z_{20809} \rtimes_{12972} Z_{51}$ (Frobenius)	bh2; mc†	master
(7, 11)	11,012,976	Haar, orbit S	$SL(2,53) \times Z_{37}$	gen + sl	master
(8, 11)	45,329,808	Haar, orbit S	SL(2,283)	gen + sl	master
(9, 11)	155,250,816	Haar, orbit S	$SL(2,223) \times Z_7$	gen + sl	master
(10, 11)	460,691,568	Haar, orbit S	SL(2,613)	gen + sl	master
(11, 11)	1,241,299,248	Haar, orbit S	SL(2,853)	gen + sl	master
(12, 11)	2,976,066,240	Haar, orbit S	$SL(2,239) \times Z_{109}$	gen + sl	master
(13, 11)	6,781,213,392	Haar, orbit S	$SL(2,563) \times Z_{19}$	gen + sl	master
(14, 11)	14,381,398,080	Haar, orbit S	$SL(2,1009) \times Z_7$	gen + sl	master
(15, 11)	28,655,908,512	Haar, orbit S	$SL(2,617) \times Z_{61}$	gen + sl (2 halves)	master
(16, 11)	29,961,590,400	Haar, orbit S (size cap)	$PSL(2,449) \times Z_{331}$	gen only	branch [d]
(3, 12)	2,368	voltage lift	cubic 4 + 4-vertex base; $Z_{37} \rtimes_6 Z_8$ = SmallGroup(296,3)	coord. rebuild + apsp	master
(4, 12)	74,066	Haar, random S	$Z_{1277} \rtimes_{964} Z_{29}$ (Frobenius)	apsp, mc; coord. apsp	master
(5, 12)	1,058,342	Haar, random S	$Z_{8969} \rtimes_{205} Z_{59}$ (Frobenius)	bh2; mc†	master
(6, 12)	9,366,240	Haar, orbit S	$PSL(2,79) \times Z_{19}$	gen + psl	master
(7, 12)	59,838,504	Haar, orbit S	$PSL(2,67) \times Z_{199}$	gen + psl	master
(8, 12)	284,095,056	Haar, orbit S	$PSL(2,167) \times Z_{61}$	gen + psl	master
(9, 12)	1,105,785,120	Haar, orbit S	$PSL(2,241) \times Z_{79}$	gen + psl	master
(10, 12)	3,712,424,352	Haar, orbit S	$PSL(2,337) \times Z_{97}$	gen + psl	master
(11, 12)	11,189,148,096	Haar, orbit S	$PSL(2,223) \times Z_{1009}$	gen + psl	master
(12, 12)	29,996,171,040	Haar, orbit S	$PSL(2,241) \times Z_{2143}$	gen + psl (2 halves)	master
(13, 12)	29,997,644,040	Haar, orbit S (size cap)	$PSL(2,181) \times Z_{5059}$	gen + psl (2 halves)	master
(14, 12)	29,996,171,040	Haar, orbit S (size cap)	$PSL(2,241) \times Z_{2143}$	gen + psl (2 halves)	master
(3, 13)	3,400	cyclic voltage lift	Desargues graph GP(10,3); Z_{170}	apsp; coord. apsp	master
(4, 13)	192,382	Haar, random S	$Z_{2237} \rtimes_{1984} Z_{43}$ (Frobenius)	apsp, mc; coord. apsp	master
(5, 13)	3,594,910	Haar, random S	$(Z_{11} \rtimes_4 Z_5) \times (Z_{2971} \rtimes_{1764} Z_{11})$, both Frobenius	mc; bh2†	branch
(6, 13)	41,734,392	Haar, orbit S	$PSL(2,83) \times Z_{73}$	gen; psl†	branch
(7, 13)	312,227,760	Haar, orbit S	$SL(2,229) \times Z_{13}$	gen; sl†	branch
(8, 13)	1,739,514,000	Haar, orbit S	$SL(2,499) \times Z_7$	gen; sl†	branch
(9, 13)	7,924,036,032	Haar, orbit S	$SL(2,593) \times Z_{19}$	gen; sl†	branch
(10, 13)	19,980,294,768	Haar, orbit S (size cap)	$PSL(2,503) \times Z_{157}$	gen only	branch [d]
(3, 14)	5,706	cyclic voltage lift	Pappus graph; Z_{317}	apsp; coord. apsp	master [e]
(4, 14)	497,042	Haar, random S	$Z_{2731} \rtimes_{2400} Z_{91}$ (Frobenius)	apsp, mc; coord. apsp	master
(5, 14)	12,795,166	Haar, random S	$Z_{148781} \rtimes_{85175} Z_{43}$ (Frobenius)	mc; bh2†	branch
(6, 14)	187,990,656	Haar, orbit S	$SL(2,97) \times Z_{103}$	gen; sl†	branch
(7, 14)	1,674,490,944	Haar, orbit S	$PSL(2,223) \times Z_{151}$	gen; psl†	branch
(8, 14)	11,325,238,056	Haar, orbit S	$PSL(2,397) \times Z_{181}$	gen; psl†	branch

Counts: 44 of the 64 values are on master, each with two independent checkers or an all-pairs BFS: 24 at D = 11, 12, the 12 Haar graphs at Δ = 17..20 and D = 8..10, 4 at D = 13, 14 with Δ = 3, 4, and (18,5), (19,5), (20,5), (17,7). For four of them, (4,11), (5,11), (6,11) and (5,12), the second checker on master's file was run by this job. 20 are on branches only: 10 at D = 13, 14 with Δ = 5..10 (nine now with two checkers, the second one run by this job; (10,13) with one), (16,11) (one checker), the four D = 6 values (exact check via automorphisms), the three level graphs at D = 7 (theorem plus sampled BFS), and two values above master's, (17,5) and (3,11), each checked all-pairs twice. One value, (3,14), is below a published census graph.

4. How the values were checked

Haar graphs. Haar(G, S) has vertices (g, A) and (g, B) for g in G and edges (g, A) ~ (sg, B) for s in S. Right multiplication by any h in G maps edges to edges and acts transitively on each side, so every vertex of one side has the same eccentricity, and the diameter is the larger of the eccentricities of (e, A) and (e, B). Two BFS therefore decide the diameter exactly. Both are needed: for odd D and non-abelian G one side can pass while the other fails (the first odd-D version of the new-cells-beyond search counted one side only; the checkers, which run both BFS, rejected a (10,11) candidate it produced, and the file was discarded). Every checker listed above builds the group from its definition (matrices mod p, or pairs (a, j) with the semidirect product rule), checks that the Δ generators are distinct, and runs both BFS. The reference checkers sl and psl were written separately from gen, with their own element indexing; above about 20 billion vertices their two BFS are run as two jobs ("2 halves"), which is the same check split in two. bh2 can also write the edge list, so small Haar graphs can be given to apsp as well.

Edge lists. The non-Haar graphs (lifts, hub graphs, the level graphs at small q) were checked by apsp, which reads an edge list and runs a BFS from every vertex (64 sources at a time) with no symmetry assumed; it also checks simplicity, the maximum degree and bipartiteness. By the lab's convention each graph is also run at diameter D − 1, where it must fail.

D = 6 (Moore graphs and Theorem D). These have 2.2–5.2 million vertices; an all-pairs BFS of each takes hours on a 4-core cloud machine (the 5.7-million-vertex (17,7) graph was checked all-pairs that way, split over eight machines). The beyond-small job checked them exactly with apsp_auto: it is given permutations of the vertex set, verifies that each maps every edge to an edge, and runs a BFS from one vertex per orbit of the group they generate (eccentricity is constant on orbits). For H(q) the permutations come from diagonal matrices that preserve the quadric and Tits' line equations (10,026 to 16,230 orbit representatives); for Theorem D at q = 17 one generator of order 16, with 200,486 orbit representatives. apsp_auto agrees with apsp on H(4) and on the Theorem D graphs at q = 5 and 7, and fails at D − 1. The H(q) values are also classical (Tits), and Theorem D is proved in Lean for every odd q.

D = 7 level graphs. (18,7), (19,7), (20,7) are 2r + 1 copies of a generalized hexagon joined by r matchings. Their diameter is at most 7 for every choice of matchings (research/synthetic/SYNTHETIC.md, Theorem 4.2), and the builder was checked all-pairs at q = 3, 4 for r = 1..3. The instances themselves (7.2–15.7 million vertices) only had a BFS from 64 random vertices, which found eccentricity 7. That is a sanity check of the builder, not a check of the graph; the value rests on the theorem. No exact check (all-pairs or by automorphisms) has been run for these three graphs.

Second checks run by this job. Where a row had only the search session's checker, this job ran a different one on the same file (sha256 in each log): sl or psl on (6,13), (7,13), (8,13), (9,13), (6,14), (7,14), (8,14); bh2 on (5,13) and (5,14); mc on master's (4,11), (5,11), (5,12) and (6,11), whose master logs have bh2 only (by the session and again by the harvest); and apsp on the 1,474-vertex (3,11) graph and the 35,728-vertex (17,5) graph. All of them reported VALID. The reference check of (10,13) (19,980,294,768 vertices) needs about 7 GB per half: the two halves did not fit in this session's memory side by side (the A half was killed after 20 minutes when the B half started), and running them one after the other would have taken about two more hours, so it was stopped unfinished. The reference check of (16,11) (29,961,590,400 vertices, about 11 GB per half) was not attempted.

5. Methods

5.1 Haar graphs with orbit-closed generator sets (Δ = 17..20 at D = 8..10; Δ ≥ 6 or 7 at D = 11..14)

Why Haar graphs, and how big. For a Haar graph the diameter condition is a covering condition in G: every element must be reached by a word of length at most D − 1 in S and S^{-1} (alternating). About $w = \Delta(\Delta - 1)^{D-2}$ such words exist. If they behaved like random elements, the expected number of elements missed would be $\mu = N\, e^{-w/N}$ with N = |G| (research/frontier/SIZE_LAW.md). Annealing closes the last holes only when μ is small (SIZE_LAW.md measured a limit μ* of about 9–40, depending on effort), so the jobs chose group orders by this "size law". The Δ = 17..20 values were taken at μ ≈ 3–13 and not pushed further; the D = 11..14 values with orbit S sit at μ ≈ 5–14 (the branch tables give μ for each row); the four cells at the size cap have μ ≈ 0.

Groups. G = SL(2, p) × Z_n or PSL(2, p) × Z_n, p prime. These have many orders near any target, cheap arithmetic, and an automorphism α of order 3: conjugation by a matrix of order 3 on the first factor, times $z \mapsto rz$ with r of order 3 mod n on the second (for n = 1, or n with no such r, α acts on the first factor only).

Orbit search. S = {e} ∪ F ∪ (orbits of α of size 3), with F a few α-fixed elements. Making S α-invariant shrinks the search space and makes the whole word set α-invariant; in the lab's earlier runs at D = 7..10 this broke the wall where random S stops working (README, research/structured/, research/theory/). The search anneals the orbit representatives; the cost is the number of uncovered elements, counted by a bit-parallel BFS (`research/bighaar/haar_big.c` on the new-cells-beyond branch, and the beyond-d11 version `research/frontier/beyond11/haar_big0.c`, 64-bit indices, N/2 bytes of memory, so 10^{10}-element groups fit in about 5 GB). beyond-d11 added two things: S made of free orbits only, without e (right for 3 | Δ), and a fix in the choice of the automorphism for orbit sizes 4, 6 and 10 (the old code could pick a matrix of order 2 or 3 there; orbits of size 3 and 5 were not affected, so no earlier result changed).

Odd D is harder than even D. For D = 2h + 1 the diameter condition is $(S^{-1}S)^h = G$ and $(SS^{-1})^h = G$. The beyond-d11 job proved (research/frontier/beyond11/ODD_D_HOLES.md, on its branch) that the words reaching an involution come in pairs (reverse and invert), and the words reaching a central element are closed under rotation by two letters, and a central hole is a hole on both sides at once. In the random model an involution is then missed with probability about $e^{-\lambda/2}$ and a central element about $e^{-\lambda/h}$, against $e^{-\lambda}$ for an ordinary element (λ = w/N). PSL(2, p) has about $p^2/2$ involutions, SL(2, p) only −I, and G × Z_n has n or 2n central elements. The rule that came out of this, and that gave most D = 11 and 13 values: at odd D use SL(2, p) × Z_n with small n; at even D, PSL(2, p) × Z_n with moderate or large n is fine. The exceptions in the table: (12,11), SL(2,239) × Z_{109}, found before the rule; (16,11), PSL(2,449) × Z_{331}, at the size cap, where λ = 41 is so large that even the special elements are reached; and at D = 13, (6,13) and (10,13), PSL × Z_n with n = 73 and 157 (beyond-d13 found that PSL with moderate n also worked there). PSL with n = 1 is the worst choice at odd D: in the measured random S almost all holes (93–98 %) were involutions, although the same groups work at even D. Measured hole counts of random S were higher than these rates predict, by 1.3–1.5 times at μ ≈ 10 and 2–2.3 times at μ ≈ 30 (beyond-d11 card r3/e14).

Size caps. The jobs set a size limit so that the low-memory checkers finish in hours: 20 billion vertices for beyond-d11 and beyond-d13, 30 billion for beyond-d11b. In (16,11), (13,12), (14,12) and (10,13) the first S tried worked at the cap, so these values are limits of checking time, not of the search. (12,12) and (15,11) are also close to the cap, but there the search had to work: μ ≈ 12 and 45 evaluations, μ ≈ 5 and 13 evaluations.

5.2 Metacyclic Haar graphs (Δ = 3..6 at D = 11..14)

At small Δ the groups are small (10^3 to 10^7 elements) and other groups can be tried. Two kinds won.

Odd-order and Frobenius groups. In $Z_m \rtimes_r Z_k$ with m prime and r of order k > 1 mod m, the centre is trivial, and if mk is odd there are no involutions (proved by both beyond-d13, `research/frontier/beyond13/THEORY.md` Fact 1, and beyond-d11b, `research/frontier/beyond11/FROBENIUS.md` and its card r2/h17). So the special holes of section 5.1 do not exist. These groups beat the SL/PSL values at odd D: (4,13) 192,382 against 170,520 for the best PSL(2,29) × Z_7 orbit graph, (5,13) 3,594,910 against 3,217,056, and (4,11) 30,334 against 24,360. Products of two such groups also work ((5,13)). The two jobs measured different

reasons. beyond-d13 points to the missing special holes. beyond-d11b (card r2/h13, at $\Delta = 4$) found that the balls of radius 8 in the Frobenius groups reach 0.99–1.00 of the random-words model and those of small SL(2, p) only 0.89–0.92, because short relations in small SL(2, p) make words collide, and that involution and central holes were not the cause in that measurement. Both effects may matter; we do not know their shares.

Random S instead of orbits. At $\Delta = 4$ the orbit form $\{e, x, \alpha x, \alpha^2 x\}$ has one free element, and at $\Delta = 5$ one fixed generator plus one orbit; a random S has $\Delta - 1$ free elements. beyond-d13 gives this as the reason the metacyclic groups with random S also won the even-D cells (4,12), (4,14) and (5,12), by 2–8 % over the orbit graphs. At (5,14) the runner-up already had random S (SL(2,97) × Z_7, 12,776,064), and the metacyclic graph won by only 0.15 %. The search is plain annealing of S (`research/frontier/haar_gen3.c`, and the beyond-d11b versions with save/resume and a targeted finisher, `research/squads/beyond11b/tools/`; beyond-d13's bit-parallel `research/frontier/beyond13/meta/hmeta.c`), plus scans of every metacyclic group in an order range with short runs on each.

Not too abelian. $Z_m \rtimes_r Z_k$ has an abelian subgroup of index k, and for small k words collide. Measured at (4,13) with random S (squad card r2/e11 of beyond-d13): the words of length 12 from (e, A) reach 1 % of G for k = 1, 6 % for k = 3, 73 % for k = 5, 91 % for k = 9 and 97–99 % for $k \geq 11$. The $\Delta = 4$, 5, 6 values in the table use single groups with k = 29 to 91, or (at (5,13)) a product of two groups with k = 5 and k = 11.

Cubic cells. At $\Delta = 3$ with $S = \{e, x, y\}$ all pairs (x, y) can be searched in each group, in 0.1–2 s per group. The new-cells-beyond job swept every metacyclic group of order 541–1,100 at D = 11 (641 groups, one r of each order per m) and found $Z_{67} \rtimes_9 Z_{11}$ (1,474 vertices), the only metacyclic hit in that range. At D = 12 the best metacyclic Haar graph (2,176, $Z_{68} \rtimes_3 Z_{16}$) is below the own-small job's voltage lift over $Z_{37} \rtimes_6 Z_8$ (2,368), which the table lists.

The metacyclic idea reached $\Delta = 6$ only at D = 11; at (6,12), (6,13) and $\Delta = 7$ it stopped at 2 to 14 uncovered elements, and SL/PSL orbit graphs stayed ahead.

5.3 Cyclic voltage lifts ($\Delta = 3$ at D = 13, 14)

Take a small cubic bipartite base graph B and a voltage x_e in Z_n on each edge; the lift has vertices (v, g) and edges $(u, g) \sim (w, g + x_e)$. Rotation $g \mapsto g + c$ is an automorphism, so only BFS from the base vertices (with g = 0) are needed. The beyond-d13 job annealed the voltages (`research/frontier/beyond13/lifts/lift.c`, from squad card r2/e16; for short runs the cooling has to be spread over about 1,000 evaluations, not the default 20,000) and then ran an exhaustive search over single-voltage changes with a tabu list (`lifts/scan_h02.c`, from card r5/h02). Results: the Desargues graph GP(10,3) over Z_{170} at D = 13 (3,400 vertices) and the Pappus graph over Z_{317} at D = 14 (5,706). Both beat every Haar graph tried at those cells (2,880 for $A_6 \times Z_4$ at D = 13, all pairs searched; 5,040 for PSL(2,7) × Z_{15} at D = 14).

An upper bound for this method (THEORY.md, Fact 2). The lift of a walk from (s, 0) ends at $(v, \Sigma\, c_e\, x_e)$, where c is the walk's net traversal vector (signed edge counts). So if the lift has diameter ≤ D, then n is at most the number of distinct net traversal vectors of walks from s to v of length ≤ D, for every pair s, v. For GP(10,3) this allows $n \leq 647$ at D = 13 and $n \leq 1{,}197$ at D = 14; the lifts found have n = 170 and (for GP(10,3) at D = 14) 282. At D = 11, 12 beyond-d13's cyclic lifts did not beat the lab's existing values, and at $\Delta = 4$, 5 lifts of $K_{4,4}$, PG(2,3), GQ(3,3) and random bases stayed below the Haar values.

5.4 Route-chart lift plus hubs: (17,5)

(17,5) was the one cell at $D \leq 7$ in this note where none of the lab's formula families applies: the multi-doubled-matching lifts need $\Delta = q + 1 +$ (an even number), and route charts need $\Delta =$ (letters) × (field size); 17 is neither for a usable prime power q. The new-cells-beyond job left only the floor 33,792, the (16,5) graph itself. The beyond-small job added vertices to that graph.

Route-chart lifts (`research/theorems/chartlift/chartlib.py`). Start from a small bipartite controller graph whose edges are properly coloured with k colours (letters); letter c carries an invertible matrix A_c and a vector b_c over F_q. Each controller vertex u becomes a fibre of vertices (u, x), x in F_q^n, and a controller edge u – v of letter c becomes the q^{n+1} edges $(u, x) \sim (v, A_c x + \beta b_c)$, β in F_q. The lift has degree kq. Along a

sequence of letters (a pattern) the reachable set in the fibre is an affine subspace; the pattern is singular if that subspace is not the whole fibre, and a good chart has few singular patterns.

Base. The (16,5) graph (`research/theorems/graphs/build_d16D5_n33792.py`): $K_{3,3}$ with a doubled perfect matching, lifted by Z_{22} (66 + 66 controller vertices), 4 letters, fibre ${F_4}^4$, with edges $((a_i, g), x) \sim ((b_j, g + v), A_c x + \beta b_c)$, β in F_4. It has 33,792 vertices, is 16-regular, has diameter 5, and every vertex has one spare degree when $\Delta = 17$.

Proposition (hubs). Let G be bipartite with sides A, B and diameter 5. Add new vertices (hubs): a hub on side A is joined to a set $T \subseteq B$, a hub on side B to a set $T \subseteq A$, each old vertex in at most one T. If (1) every vertex of the hub's own side is within distance 3 of T, and (2) any two hubs on the same side have neighbours x, x′ with $d_G(x, x') \le 2$, then the new graph has diameter ≤ 5.

Proof. Old pairs are still within 5. A hub h on side A reaches all of A within 4 by (1), and all of B within 5, since distances between B-vertices in G are at most 4 (G is bipartite with diameter 5). Two A-hubs are within 4 by (2). An A-hub reaches every A-vertex within 4, so it reaches every B-hub, which has a neighbour in A, within 5. □

The two sides are independent, so the best packings of each side combine. With $|T| = 16$ the maximum degree is 17.

Hubs from fibre lines, in orbits. A set T of 4 affine lines $\{(v, y + \beta c) : \beta \in F_4\}$ in 4 controller vertices covers a whole side within distance 3 for many choices (squad card r1/c03). Translation $g \mapsto g + 1$ on the controller coordinate is an automorphism of the base, so a good T gives 22 hubs at once (card r2/d06), provided its translates are disjoint and pairwise compatible. Call the orbit of an old vertex under the translation its key; a side has 16,896 / 22 = 768 keys, the translates of T are disjoint exactly when its 16 vertices have distinct keys, and orbits must not share keys. So a side holds at most 48 orbits, and the method is capped at $33{,}792 + 2 \cdot 48 \cdot 22 = 35{,}904$ vertices.

Packing. Greedy growth of orbit bases saturated at 32 orbits on side A and 37 on side B. Destroy-and-repair loops (drop 2–5 orbits, regrow for 60 s, keep if not worse), an annealer over orbit packings, and "region" destroys (drop every orbit meeting a random affine set of keys) brought side A to 43 orbits and side B to 45: 88 orbits, 1,936 hubs, 35,728 vertices, 25.6 % of the Moore bound. Twelve more search cards in a fifth round found no 44th or 46th orbit. Master has the earlier 35,618 (83 orbits), checked all-pairs by the coordinator; 35,728 is checked all-pairs by the session, by a separate plain BFS from every vertex (`bfs_all`), and by this job.

5.5 The other rows

- **(18,5), (19,5): doubled-matching lifts of W(q)** (`research/frontier/beyond/MDM.md` on the branches, `research/d10D7/DM_D5.md`). With k doubled matchings over $Z_2 \times Z_{2k+1}$ the lift of the symplectic quadrangle W(q) has degree $q + 1 + 2k$ and $4(2k + 1)(q + 1)(q^2 + 1)$ vertices, and diameter ≤ 5 as soon as one labelling of the flags by parities splits the $q + 1$ shortest paths between every two opposite elements (elements at distance 4 in W(q)); the matchings can be anything. (18,5): $q = 13$, $k = 2$; (19,5): $q = 16$, $k = 1$ (Z_6).
- **(20,5): route-chart lift over F_5.** The (16,5) construction with fibre ${F_5}^4$ and 4 letters × 5 scalars, controller $K_{3,3}$ + doubled matching lifted by Z_{20} (60 controller vertices per side).
- **D = 6.** H(q) for q = 16, 17, 19 are Moore graphs; (19,6) is Theorem D of the lab's polished notes (the hexagon H(17) doubled with levels and an exact whisker set), proved for odd q and checked in Lean.
- **D = 7.** 2r + 1 copies of H(q) joined by r matchings (section 4); for (17,7) the conics-2 job freed some incidences of the 7-copy cover of H(13) and used the freed slots for 32,356 + 30,327 hub vertices joined by a hub graph of degree at most 10 (SYNTHETIC.md §4.2), 5,693,959 against 5,631,276 for the plain cover. For (18,7), (19,7), (20,7) the plain covers are listed: the new-cells-beyond job tried a simpler hub surgery, did not establish its hub condition for q = 13 and 16, and estimated its gain at 0.1–0.4 %; the conics-2 construction (which gained 1.1 % at (17,7)) was run for (17,7) only.

6. What failed

- **(17,5) before the hubs.** A fifth translation vector on one route-chart letter (17 = 5 + 4 + 4 + 4) cut the far pairs at Z_{23} from 106,000 to 47,000 but not to 0, and cannot repair a singular pattern (section 5.4) that uses the extended letter once. Lifts of W(13) with 3 extra degrees (8 sheets): with a doubled and a plain matching, the director's argument suggests that the conditions (needed for almost all pairs) force the labelling to be a coboundary, which kills the needed parity condition unless the matching is a polarity (a duality of order 2); on W(8), which has one, the far pairs fell from 174,000 to 2,432 but not to 0. In any cover of W(16) two copies of one vertex are at distance more than 5 (a path between them projects to a closed walk with nonzero voltage, which contains a cycle of W(16), and W(16) has girth 8). Haar graphs: best 20,200 (AGL(1,101)). A projective-plane scaffold for the hubs left 12,182 far pairs. Hub packing itself sits on a plateau at 43 + 45 orbits.
- **Larger μ at D = 11.** (12,11) SL(2,601) × Z_7 (μ 10.7) stopped at 2 holes after about 230 evaluations and SL(2,193) × Z_{211} (μ 10.3) at 2 with −I its only hole; (11,11) PSL(2,257) × Z_{73} (μ 10.1) stopped at 9, while SL(2,853) at the same μ solved after 892 evaluations. The later job beyond-d11b did not try to raise (11..14, 11) or (11,12): the gains would be 1–3 % and its machine was busy with the 30-billion-vertex checks.
- **Near misses that did not close** (1–3 uncovered elements): (6,11) SL(2,103), (7,11) SL(2,67) × Z_{19}, (9,11) SL(2,137) × Z_{31}, (6,12) PSL(2,211), (9,12) PSL(2,167) × Z_{241}, (6,13) SL(2,59) × Z_{103} (the two holes are a transpose pair), (6,14) PSL(2,83) × Z_{331}, (7,13) PSL(2,131) × Z_{139}; and (9,13) SL(2,677) × Z_{13} stopped at 12.
- **Other shapes for S.** Orbits of size 2, 4, 5, 6 at D = 11; transpose-closed S; −I as a generator; orbit-closed S inside metacyclic groups; a meet-in-the-middle finisher. None beat the order-3 orbits (in SL/PSL) or random S (in metacyclic groups). Near the limit the annealer's 1- and 2-hole states seem to be true local minima: for one 1-hole (4,12) state all 105,609 single-generator swaps were tried and none closed it.
- **Other groups and bases.** Metacyclic groups at Δ ≥ 6 (except (6,11)); Frobenius groups at (7,11) and (8,11) (2 and 8 holes); non-abelian voltage groups on GP(10,3) (2 holes at 3,300 vertices); in beyond-d11b's first round Sz(8), $F{\sim}p^3 \rtimes C_k$, bipartite Cayley graphs of PGL(2,p), PSL(2,p^e) with a field automorphism, AGL(1,q) products, and voltage lifts of PG(2,3) (19,500 at (4,11)).
- **D = 8 from hexagons.** The 2013 table's D = 8 values follow a pattern $(2j + 1)(\lfloor (q - 1)/2 \rfloor + 3(j - 1)) \cdot |H(q)|$ (`research/theorems/REGULARITIES.md`). Its j = 2 case is the order of Gómez's compound graphs ΩH_q, which as defined have diameter at least 9 (research/synthetic/SYNTHETIC.md §6; a diameter-8 version with a labelling for each copy is known only for q = 2, 3); for other j no construction is known. Card r1/c10 of new-cells-beyond found a Z_{20} lift of H(3) plus two matchings with diameter 8 (14,560, the pattern's value at q = 3, j = 2) by annealing, but q = 4 stalled at 150 holes. If the pattern could be realized for large q, it would give about 1.6–1.9 times the Haar values at D = 8 (for example about $2.2 \cdot 10^8$ at (20,8) against $1.2 \cdot 10^8$).
- **D = 5 otherwise.** Route charts over F_3 (6 letters) and F_2 (9 letters) reached only 29,160 at (18,5) in 25-minute runs. Blind annealing of voltages on W(13) stayed in the millions of uncovered pairs where the structured doubled-matching lift solves in seconds.
- **The machines.** Four cores and 15 GB per cloud session. A 20-billion-vertex check was killed once for memory when a squad card's checker and a search ran beside it; squad cards were then capped at 0.5 GB, which also capped what they could search at (7,14) and (8,13). Checking time grows with p as well as N (haarcheck_gen's field tables leave the L2 cache near p ≈ 850–1,000): the 14.4-billion-vertex (14,11) check took about 4.4 CPU-hours.

7. Loose ends

1. The reference check of (16,11) and (10,13): `haarcheck_psl FILE A` and `haarcheck_psl FILE B` (research/structured/tools/haarcheck_psl.c) on `d16D11_psl_449_331_orb3-5-1_s1.txt` (beyond-d11b branch) and `d10D13_psl_503_157_orb3-3-1_s1.txt` (beyond-d13 branch), about 11 GB and 7 GB per half.
2. An exact check of the three level graphs at D = 7, or the hub step of (17,7) applied to them.
3. The coordinator's own check and recording of the 20 branch-only values: (17,5) 35,728 and (3,11) 1,474, which are above master's values, the D = 13, 14 values at Δ = 5..10, (16,11), and the four D =

6 values.

4. An apsp check of the census graph CSS(6272,7) at (3,14) (and CAT(3360,4) at (3,13)), and a literature check for all 64 cells.

8. Files

- Table rows and PDF: `research/paper/make_beyond.py` (re-derives every vertex count and percentage and rewrites the table in BEYOND.md; `--pdf` also runs pandoc and tectonic). This note: `research/paper/BEYOND.md`, `BEYOND.tex`, `BEYOND.pdf`.
- Second checks of this job: `research/paper/beyond_checks/` (README there).
- Haar solution files: `research/frontier/sols/d<Δ>D<D>_*.txt` (master for D ≤ 12 except (16,11); the beyond-d13 branch for D = 13, 14 (and `research/frontier/beyond13/sols/` there for the metacyclic (4,12) 74,066, of which master has only the edge list); the beyond-d11b branch for (16,11)). Rebuild and check one with `python3 research/frontier/build_haar2.py FILE` (small) or `cc -O3 -o sl research/frontier/haarcheck_sl.c && ./sl FILE`.
- Edge lists: `research/frontier/beyond/graphs/` (master and branches), `research/frontier/beyond13/graphs/` (beyond-d13), `research/squads/beyond-small/graphs/` (beyond-small); the files are gzipped; check with `cc -O3 -o apsp research/lifts/coordinator_check/apsp.c && gunzip -k G.edges.gz && ./apsp G.edges Δ D`.
- Squad logs: `research/squads/LOG.md` and `research/squads/beyond*/` on each branch.